

Lecture 19/Module 2

Transient Analysis of First and Second Order Circuit

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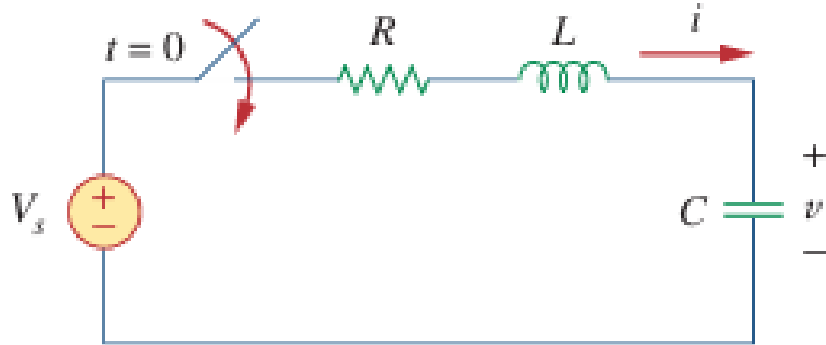


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Today's Topics

- Step Response of Series RLC Circuit
- Step Response of Parallel RLC Circuit

Step response of series RLC circuit



The step response is obtained by the sudden application of a dc source.

Consider no initial current through the inductor and initial voltage across the capacitor. The current through the inductor and initial voltage across the capacitor just after the switch is closed is the same as the initial condition.

Applying KVL around the loop for $t < 0$,

$$L \frac{di}{dt} + Ri + v = V_s \quad \text{--- 1}$$

But,

$$i = C \frac{dv}{dt} \quad \text{--- 2}$$

Substituting for i and rearranging,

$$\frac{d^2v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{v}{LC} = \frac{V_s}{LC} \quad \text{--- 3}$$

The solution to Eqn. 3 has two components: the transient response $v_t(t)$ and the steady-state response $v_{ss}(t)$ that is,

$$v(t) = v_t(t) + v_{ss}(t)$$

Step response of series RLC circuit

$$v_t(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (\text{Overdamped})$$

$$v_t(t) = (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically damped})$$

$$v_t(t) = (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Underdamped})$$

--- 4

The steady-state response is the final value of $v(t)$. The final value of the capacitor voltage is the same as the source voltage V_s .

$$v_{ss}(t) = v(\infty) = V_s \quad \text{--- 5}$$

The complete solutions for the overdamped, underdamped, and critically damped cases are:

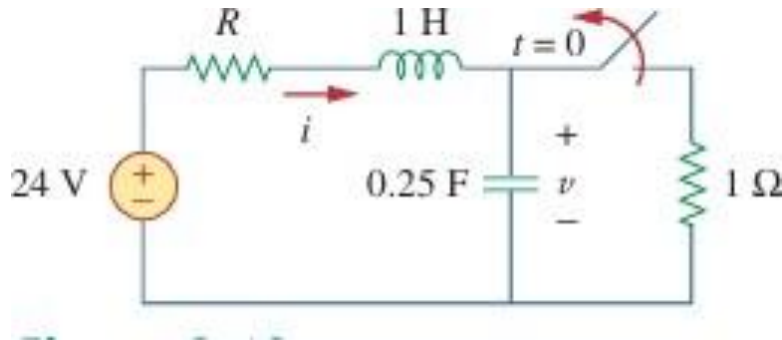
$$v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (\text{Overdamped})$$

$$v(t) = V_s + (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically damped})$$

$$v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Underdamped})$$

Exercise

For the circuit, find $v(t)$ and $i(t)$ for $t > 0$. Consider the cases when $R = 5\ \Omega$, $R = 4\ \Omega$, and $R = 1\ \Omega$.



when $R = 5\ \Omega$, $t < 0$

the switch is closed for a long time. The capacitor behaves like an open circuit while the inductor acts like a short circuit. The initial current through the inductor is:

$$i(0) = \frac{24}{5 + 1} = 4\text{ A} \quad \text{--- 1}$$

The initial voltage across the capacitor is the same as the voltage across the $1\ \Omega$ resistor; that is,

$$v(0) = 1i(0) = 4\text{ V} \quad \text{--- 2}$$

Exercise

For $t > 0$, the switch is opened, so that we have the $1\ \Omega$ resistor disconnected. What remains is the series RLC circuit with the voltage source. The characteristic roots are determined as follows:

$$\alpha = \frac{R}{2L} = \frac{5}{2 \times 1} = 2.5, \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 0.25}} = 2$$
$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -1, -4 \quad \text{--- 3}$$

We find that $\alpha > \omega_0$ – we have an overdamped natural response. The total response is :

$$v(t) = v_{ss} + (A_1 e^{-t} + A_2 e^{-4t})$$

Where v_{ss} is the steady-state response. It is the final value of the capacitor voltage and $v_f = 24\text{ V}$

$$v(t) = 24 + (A_1 e^{-t} + A_2 e^{-4t}) \quad \text{--- 4}$$

We now need to find and using the initial conditions.

$$v(0) = 4 = 24 + A_1 + A_2$$

$$-20 = A_1 + A_2 \quad \text{--- 5}$$

Exercise

The current through the inductor cannot change abruptly and is the same current through the capacitor at $t = 0^+$ because the inductor and capacitor are now in series. Hence,

$$i(0) = C \frac{dv(0)}{dt} = 4 \quad \Rightarrow \quad \frac{dv(0)}{dt} = \frac{4}{C} = \frac{4}{0.25} = 16 \quad \text{--- 6}$$

Taking derivative of v in eqn.4,
$$\frac{dv}{dt} = -A_1 e^{-t} - 4A_2 e^{-4t} \quad \text{--- 7}$$

at $t = 0$,
$$\frac{dv(0)}{dt} = 16 = -A_1 - 4A_2 \quad \text{--- 8}$$

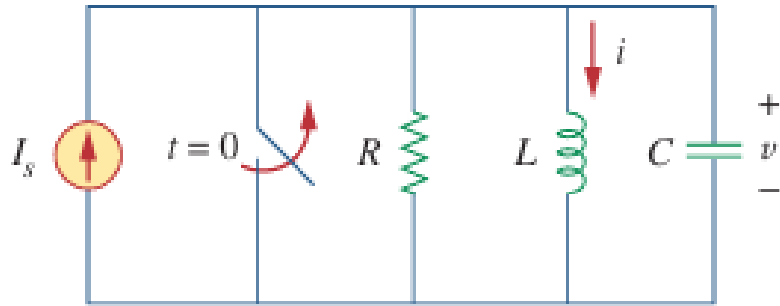
From eqn. 5 and eqn.8,
$$A_1 = -64/3 \text{ and } A_2 = 4/3.$$

eqn. 4 becomes,
$$v(t) = 24 + \frac{4}{3}(-16e^{-t} + e^{-4t}) \text{ V} \quad \text{--- 9}$$

Since the inductor and capacitor are in series for $t > 0$, the inductor current is the same as the capacitor current.
$$i(t) = C \frac{dv}{dt}$$

Multiplying eqn. 7 by $C = 0.25$ and substituting the values of A_1 and A_2 ,
$$i(t) = \frac{4}{3}(4e^{-t} - e^{-4t}) \text{ A}$$

Step response of parallel RLC circuit



Applying KCL at the top node for $t > 0$

$$\frac{v}{R} + i + C \frac{dv}{dt} = I_s \quad \text{--- 1}$$

$$v = L \frac{di}{dt}$$

$$\frac{d^2 i}{dt^2} + \frac{1}{RC} \frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC} \quad \text{--- 2}$$

The complete solution to Eqn.2 consists of the transient response $i_t(t)$ and the steady-state response $i_{ss}(t)$

$$i(t) = i_t(t) + i_{ss}(t) \quad \text{--- 3}$$

Step response of parallel RLC circuit

$$i(t) = I_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (\text{Overdamped})$$

$$i(t) = I_s + (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically damped})$$

$$i(t) = I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Underdamped})$$

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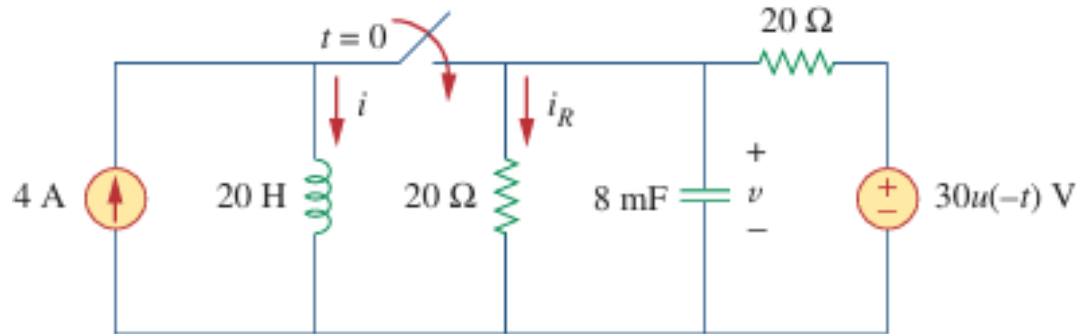
The constants and in each case can be determined from the initial conditions for i and di/dt .

The complete response for any variable may be found directly, using

$$x(t) = x_{ss}(t) + x_t(t)$$

Exercise

In the circuit, find $i(t)$ and $i_R(t)$ for $t > 0$.



For $t < 0$, the switch is open, and the circuit is partitioned into two independent sub-circuits. The 4-A current flows through the inductor

$$i(0) = 4 \text{ A} \quad \text{--- 1}$$

$30 u(-t) = 30 \text{ V}$ when $t < 0$, and *equal to 0* when $t > 0$

$$v(0) = \frac{20}{20 + 20}(30) = 15 \text{ V} \quad \text{--- 2}$$

Exercise

For $t > 0$, the switch is closed, and we have a parallel RLC circuit with a current source. The voltage source is zero which means it acts like a short-circuit. The two resistors are now in parallel.

$$\begin{aligned}\alpha &= \frac{1}{2RC} = \frac{1}{2 \times 10 \times 8 \times 10^{-3}} = 6.25 \\ \omega_0 &= \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{20 \times 8 \times 10^{-3}}} = 2.5 \\ s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -6.25 \pm \sqrt{39.0625 - 6.25} \\ &= -6.25 \pm 5.7282 \\ s_1 &= -11.978, \quad s_2 = -0.5218 \quad \text{--- 3}\end{aligned}$$

Since $\alpha > \omega_0$ we have the over damped case.

$$i(t) = I_s + A_1 e^{-11.978t} + A_2 e^{-0.5218t} \quad \text{--- 4}$$

I_s the final value of $i(t)$ which is equal to 4 A .

Use the initial conditions to determine A_1 and A_2 .

Exercise

$$i(0) = 4 = 4 + A_1 + A_2 \quad \Rightarrow \quad A_2 = -A_1$$

$$\frac{di}{dt} = -11.978A_1e^{-11.978t} - 0.5218A_2e^{-0.5218t}$$

At $t = 0$

$$\frac{di(0)}{dt} = -11.978A_1 - 0.5218A_2 \quad \text{--- 5}$$

But,

$$L \frac{di(0)}{dt} = v(0) = 15 \quad \Rightarrow \quad \frac{di(0)}{dt} = \frac{15}{L} = \frac{15}{20} = 0.75$$

Applying this in Eqn.5, we get

$$0.75 = (11.978 - 0.5218)A_2 \quad \Rightarrow \quad A_2 = 0.0655$$

Thus, $A_1 = -0.0655$ and $A_2 = 0.0655$

$$i(t) = 4 + 0.0655(e^{-0.5218t} - e^{-11.978t}) \text{ A}$$

$$i_R(t) = \frac{v(t)}{20} = \frac{L}{20} \frac{di}{dt} = 0.785e^{-11.978t} - 0.0342e^{-0.5218t} \text{ A}$$

Thank You